

UTSAB PANDEY

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Authorized to work in the U.S. via CPT — no sponsorship required from employer

EDUCATION

Texas State University — B.S. Mechanical Engineering

Expected Dec 2028

GPA: 4.0/4.0 | Honors College Scholar | President's Honors Scholarship | President's List (Spring & Fall 2025)

Honor Society: Alpha Lambda Delta

Organizations: Bobcat Aerospace · RAS IEEE (Robotics and Automation Society) — building a sumo bot

TECHNICAL SKILLS

CAD / CAE: SolidWorks (Thermal & Structural FEA), AutoCAD, Fusion 360, Simulink

Engineering Methods: GD&T, DFM, tolerance analysis, FMEA, BOM review, technical documentation, engineering report writing

Programming: MATLAB (Certified), Python, Arduino, data acquisition systems

Fabrication & Hands-On: Gas & arc welding, soldering, milling, 3D printing, thin-film deposition, epoxy masking, optical fiber coating

Optical Lab Equipment: Fiber manipulators, optical circulators, fusion splicers, fiber cleavers, power meters

Measurement & Lab Practice: Atomic Force Microscopy (AFM), oscilloscopes, surface characterization, nanoscale measurement, comfortable handling lab chemicals per protocol

Software: Microsoft Excel, Teams, Office Suite

Certifications: AutoCAD (2025), MATLAB (2025), Fusion 360 (2025), Arduino Foundations (2025)

RESEARCH EXPERIENCE

Undergraduate Researcher — AFM Parametric Sensing Lab

Oct 2025 – Present

Texas State University — NSF/NIH/USDA Funded (Oct 2025 – Oct 2029)

- Engineered precision TiO₂-coated optical fiber electrodes for AFM cantilever actuation via thin-film deposition, using a fiber manipulator and 1550 nm optical circulator to route real-time power meter feedback, targeting ~30% reflectivity for reliable opto-mechanical sensing.
- Reduced coating rework to 1 attempt per sample by developing a selective epoxy masking method for localized thin-film deposition on sub-millimeter fiber geometries, improving throughput and sample consistency.
- Produced uniform Ag thin films via chemical silver deposition (AgNO₃ + glucose reduction), handling reagents per lab safety protocol; characterized surface morphology using AFM to support parametric stiffness modulation (2f₀) cantilever control experiments.
- Achieved 0.01 dB fusion splice loss on sub-millimeter optical fibers via a precision cleave-and-align protocol; presented this research at the Texas State STEM Conference and to visiting UT Austin students during PREM Day.

ENGINEERING PROJECTS

Biomass Pyrolysis Reactor — Full Design and Thermal Simulation | SolidWorks

2025 – 2026

- Designed industrial-scale reactor geometry in SolidWorks for 1000°C biomass pyrolysis, applying heat transfer principles, DFM guidelines, and high-temperature alloy selection with pressure containment systems.
- Validated double-jacket thermal isolation via steady-state FEA (800 W, 700°C inner jacket / 25°C outer); confirmed 986 K peak at inner wall with outer jacket near ambient, quantifying insulation performance.
- Produced a professional engineering report documenting full simulation methodology, Stefan-Boltzmann radiation analysis, material justification, and results interpretation.

ATM Accessibility Mechanical Prototype | SolidWorks

2025

- Designed and fabricated a linkage-based interface mechanism to improve ATM usability for users with physical limitations; identified a failure mode from excessive activation force, redesigned lever geometry to reduce friction, and validated via 3D-printed prototype testing.

LEADERSHIP & COMMUNICATION

Supplemental Instruction Leader — Engineering & General Chemistry

Fall 2025 – May 2026

Texas State University

- Designed and led structured review sessions for 90+ engineering students per semester covering thermodynamics, kinetics, and energy systems; adapted delivery in real time based on student comprehension.

Autodesk Design & Make Ambassador

2025 – Present

Autodesk

- Delivered hands-on CAD workshops in Fusion 360, AutoCAD, parametric modeling, and stress analysis to engineering students campus-wide as Autodesk's official academic representative at Texas State University.